Mastery in Data Analytics and Visualizations and Career Advancement

Lab sheet 01

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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import MinMaxScaler, StandardScaler

# 1. Load dataset
df = pd.read_csv("titanic.csv")

# 2. Display dataset
print("First 5 Records:")
print(df.head())

print("\nDataset Information:")
df.info()

print("\nStatistical Description:")
print(df.describe())

# 3. Check missing values
print("\nMissing Values:")
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print(df.isnull().sum())

# 4. Handle missing values
if "Age" in df.columns:
    df["Age"] = df["Age"].fillna(df["Age"].median())

if "Embarked" in df.columns:
    df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])

if "Fare" in df.columns:
    df["Fare"] = df["Fare"].fillna(df["Fare"].median())

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# 5. Remove duplicate records
print("\nDuplicate Records:", df.duplicated().sum())
df = df.drop_duplicates()

# 6. Standardize categorical values
if "Sex" in df.columns:
    df["Sex"] = df["Sex"].str.lower().str.strip()

if "Embarked" in df.columns:
    df["Embarked"] = df["Embarked"].str.upper().str.strip()

# 7. One-Hot Encoding
categorical_columns = []
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for col in ["Sex", "Embarked"]:
    if col in df.columns:
        categorical_columns.append(col)

if categorical_columns:
    df = pd.get_dummies(
        df,
        columns=categorical_columns,
        drop_first=True
    )

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# 8. Detect outliers using IQR
if "Fare" in df.columns:
            Q1 = df["Fare"].quantile(0.25)
    Q3 = df["Fare"].quantile(0.75)
    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    df["Fare"] = df["Fare"].clip(lower, upper)

# 9. Feature Engineering
if "SibSp" in df.columns and "Parch" in df.columns:
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    df["FamilySize"] = df["SibSp"] + df["Parch"] + 1

if "Age" in df.columns:
    df["AgeGroup"] = pd.cut(
        df["Age"],
        bins=[0, 12, 18, 60, np.inf],
        labels=["Child", "Teenager", "Adult", "Senior"]
    )
# Convert AgeGroup to numerical form
if "AgeGroup" in df.columns:
    df["AgeGroup"] = df["AgeGroup"].cat.codes

# 10. Normalize numerical attributes
if "Fare" in df.columns:
    scaler = MinMaxScaler()
    df["Fare"] = scaler.fit_transform(df[["Fare"]])

# 11. Check cleaned data
print("\nCleaned Dataset:")
print(df.head())
print("\nRemaining Missing Values:")
print(df.isnull().sum())

# 12. Visualize an attribute
if "Age" in df.columns:
    plt.figure(figsize=(8, 5))
    sns.boxplot(x=df["Age"])
    plt.title("Age Distribution After Preprocessing")
```

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    plt.show()
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# 13. Save cleaned dataset
df.to_csv("cleaned_titanic.csv")
print("\nCleaned dataset saved successfully.")

# Q1. Why is data preprocessing considered one of the most important phases in data analytics?

# Data preprocessing is important because real-world data may contain missing values, duplicate records, inconsistent formats, categorical variables, and outliers. Cleaning and transforming the data improves its quality and reliability before analysis, visualization, or machine learning.

# Q2. Explain different methods of handling missing values with suitable examples.
# Missing values can be handled using several methods:

# Mean: Replace missing numerical values with the average value.
# Median: Replace missing numerical values with the middle value. It is useful when data contains outliers.
# Mode: Replace missing categorical values with the most frequently occurring value.
# Row removal: Remove rows containing missing values.
# Column removal: Remove a column if a very large amount of its data is missing.

# For example, missing Age values in the Titanic dataset can be replaced with the median age.
```

Q3. Differentiate between Label Encoding and One-Hot Encoding.

Label Encoding One-Hot Encoding

Converts categories into numerical labels. Creates separate columns for each category.

Example: Male = 0, Female = 1. Example: Male and Female become separate binary columns.

Uses one column. Usually creates multiple columns.

Suitable when categories have an order or when numeric labels are appropriate. Useful for nominal categories without an inherent order.

The experiment identifies both Label Encoding and One-Hot Encoding as methods for converting categorical variables into numerical form.

Q4. What are outliers? How can they affect analytical results?

Outliers are observations that are unusually high or low compared with most other observations.

They can:

Distort the mean.

Increase standard deviation.

Affect correlations.

Influence statistical analysis.

Reduce the performance of some machine learning models.

Outliers can be detected using box plots and the IQR method.

Q5. Explain the difference between normalization and standardization.

Normalization transforms values into a fixed range, commonly between 0 and 1.

Formula:

X' = (X - Xmin) / (Xmax - Xmin)

Standardization transforms data so that it has a mean of 0 and a standard deviation of 1.

Formula:

Z = (X - Mean) / Standard Deviation

Normalization is useful when a fixed range is required, while standardization is useful when algorithms work better with centered and scaled data.

The experiment requires numerical attributes to be normalized or standardized to bring them to a common scale.

Q6. Why should duplicate records be removed before analysis?

Duplicate records should be removed because they represent repeated observations. If they are not removed, they can:

Increase the frequency of certain observations.

Produce biased statistics.

Affect averages and other calculations.

Influence machine learning models.

Reduce data consistency.

Therefore, duplicate records should be removed before analysis.

Q7. What is feature engineering? Give two practical examples.

Feature engineering is the process of creating new useful attributes from existing data.

Example 1: Family Size

df["FamilySize"] = df["SibSp"] + df["Parch"] + 1

Example 2: Age Grou

Age can be divided into categories such as:

Child

Teenager

Adult

Senior

These examples are consistent with the experiment's suggested Family Size and Age Group features.

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Q8. Which preprocessing techniques would you apply to the IBM HR Employee Attrition dataset and why?

The following techniques can be applied:

Missing-value handling — to deal with incomplete records.

Duplicate removal — to prevent repeated employee records.

Categorical encoding — to convert attributes such as department or job role into numerical form.

Outlier detection — to identify unusually high or low numerical values such as income.

Normalization/standardization — to bring numerical attributes to a common scale.

Feature engineering — to create useful attributes such as income categories or age groups.

These techniques correspond to the preprocessing methods required by the experiment.

Q9. How does poor-quality data affect machine learning model performance?

Poor-quality data can negatively affect machine learning models. Missing, incorrect, duplicate, inconsistent, and noisy data can cause the model to learn incorrect patterns.

As a result:

Prediction accuracy may decrease.

Model performance may become unreliable.

Results may become biased.

The model may fail to generalize to new data.

Therefore, data should be properly cleaned and preprocessed before machine learning.

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Q10. Name any three Python libraries commonly used for data preprocessing.

Three commonly used Python libraries are:

Pandas — data loading, cleaning, and manipulation.

NumPy — numerical calculations and array operations.

Scikit-learn — preprocessing, scaling, encoding, and machine learning.

The experiment specifically lists Pandas and NumPy among its required libraries and also specifies Matplotlib and Seaborn.

OUTPUT

```
PS D:\PROGRAMS\Career advancement lab\lab sheet 1> python -u "d:\PROGRAMS\Career advancement lab\lab sheet 1\expe.py"
First 5 Records:
   PassengerId  Survived  Pclass                Name     Sex   Age  SibSp  Parch   Fare Embarked
0            1         0       3      Braund, Mr. Owen    male  22.0      1      0   7.25        S
1            2         1       1    Cumings, Mrs. John  female  38.0      1      0  71.28        C
2            3         1       3   Heikkinen, Miss Laina  female  26.0      0      0   7.92        S
3            4         1       1  Futrelle, Mrs. Jacques  female   NaN      1      0  53.10        S
4            5         0       3     Allen, Mr. William    male  35.0      0      0   8.05        S

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 10 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  21 non-null     int64
 1   Survived     21 non-null     int64
 2   Pclass       21 non-null     int64
 3   Name         21 non-null     str
 4   Sex          21 non-null     str
 5   Age          18 non-null     float64
 6   SibSp        21 non-null     int64
 7   Parch        21 non-null     int64
 8   Fare         21 non-null     float64
```

9 Embarked 21 non-null str

dtypes: float64(2), int64(5), str(3)

memory usage: 2.3 KB

Statistical Description:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	21.000000	21.000000	21.000000	18.000000	21.000000	21.000000	21.000000
mean	10.047619	0.476190	2.238095	28.777778	0.619048	0.190476	33.668571
std	6.127611	0.511766	0.889087	14.031710	1.071270	0.511766	47.607696
min	1.000000	0.000000	1.000000	2.000000	0.000000	0.000000	7.250000
25%	5.000000	0.000000	1.000000	22.000000	0.000000	0.000000	8.050000
50%	10.000000	0.000000	3.000000	28.000000	0.000000	0.000000	13.000000
75%	15.000000	1.000000	3.000000	37.250000	1.000000	0.000000	30.070000
max	20.000000	1.000000	3.000000	54.000000	4.000000	2.000000	211.000000

Missing Values:

PassengerId 0

Survived 0

Pclass 0

Name 0

Sex 0

Age 3

SibSp 0

Parch 0

Fare 0

Embarked 0

dtype: int64

Duplicate Records: 0

Cleaned Dataset:

PassengerId Survived Pclass Name Age SibSp Parch Fare Sex_male Embarked_Q Embarked_S FamilySize AgeGroup

0 1 0 3 Braund, Mr. Owen 22.0 1 0 0.000000 True False True 2 2

1 2 1 1 Cumings, Mrs. John 38.0 1 0 1.000000 False False False 2 2

2 3 1 3 Heikkinen, Miss Laina 26.0 0 0 0.011996 False False True 1 2

3 4 1 1 Futrelle, Mrs. Jacques 28.0 1 0 0.820949 False False True 2 2

4 5 0 3 Allen, Mr. William 35.0 0 0 0.014324 True False True 1 2

Remaining Missing Values:

PassengerId 0

Survived 0

Pclass 0

Name 0

Age 0

SibSp 0

Parch 0

Fare 0

Sex_male 0

Embarked_Q 0

Embarked_S 0

FamilySize 0

AgeGroup 0

dtype: int64

Cleaned dataset saved successfully.

